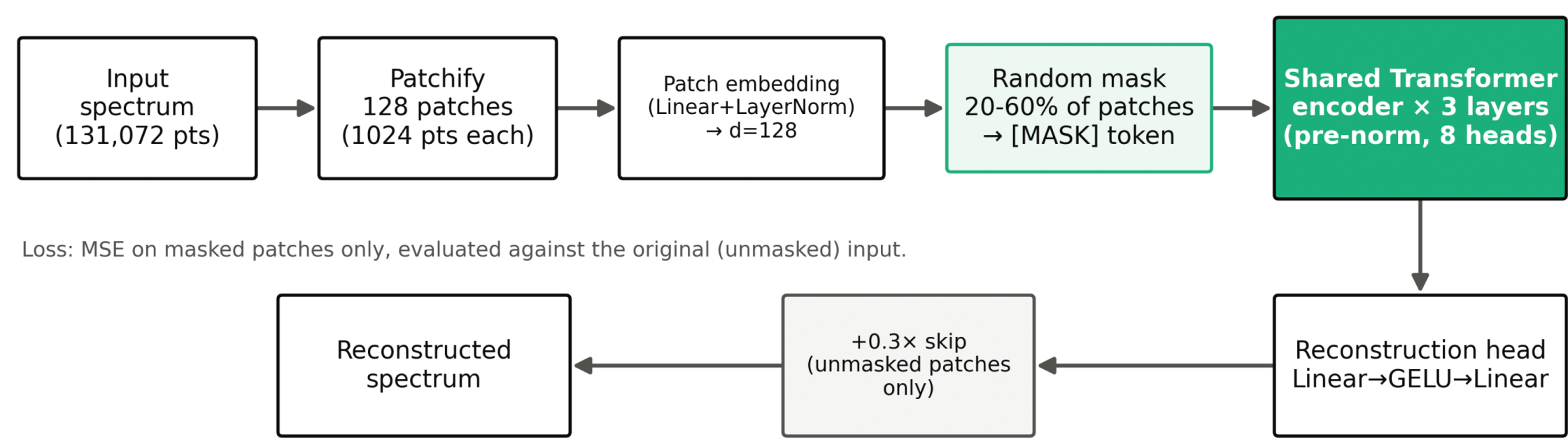


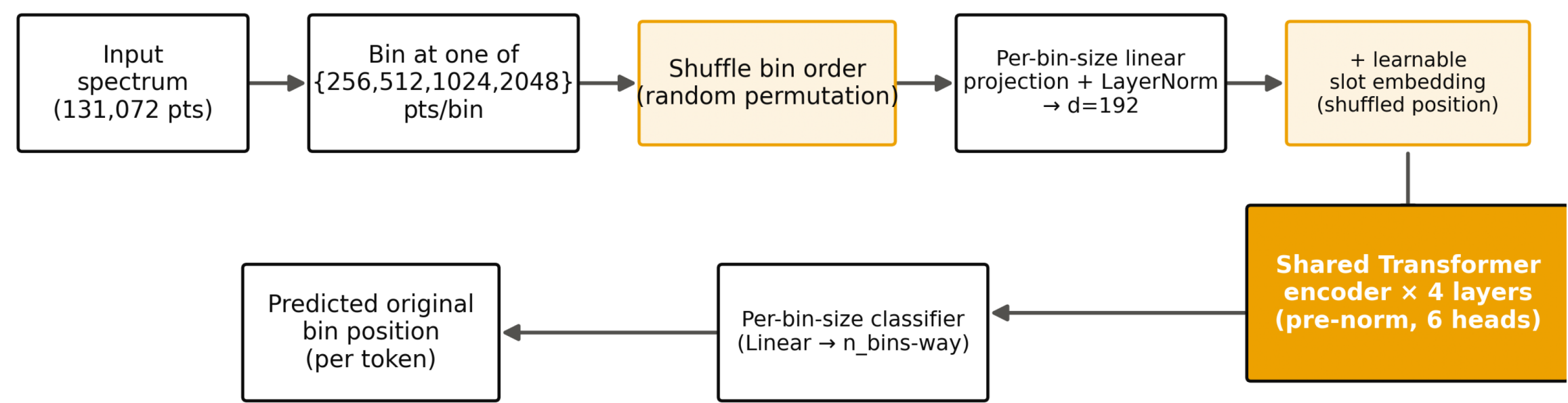
Masked-reconstruction SSL

trainer_revised.NMRMaskedAutoencoder -- patch=1024, d_model=128, 3 layers, 8 heads, FFN=256



Jigsaw (multibin) SSL

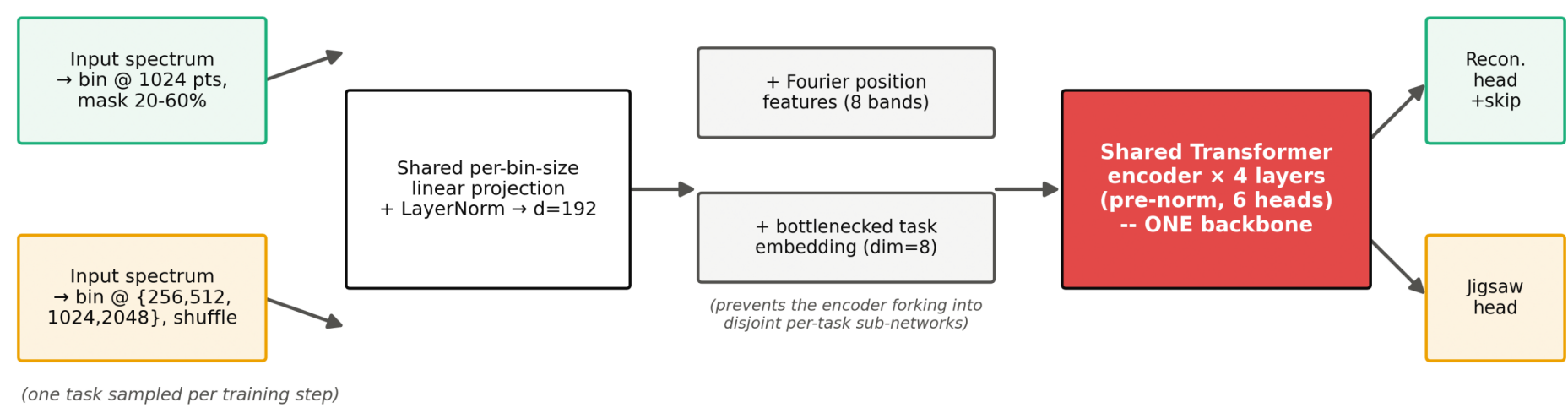
train_jigsaw_spectra.JigsawNMRModel -- bin sizes {256,512,1024,2048}, d_model=192, 4 layers, 6 heads, FFN=768



Loss: cross-entropy between predicted and true original bin index (label smoothing 0.05); trained across all 4 bin sizes.

Joint masked + jigsaw SSL

train_joint_ssl.JointNMRSSLModel -- shared d_model=192, 4 layers, 6 heads, FFN=768, task_embed_dim=8 (bottlenecked), Fourier position bands=8



Masked loss: MSE over only the top 17.5% highest-magnitude bins per spectrum (peak_top_fraction) -- avoids the loss being trivially satisfied by predicting a flat baseline. Jigsaw loss: cross-entropy, same as the standalone jigsaw model.

